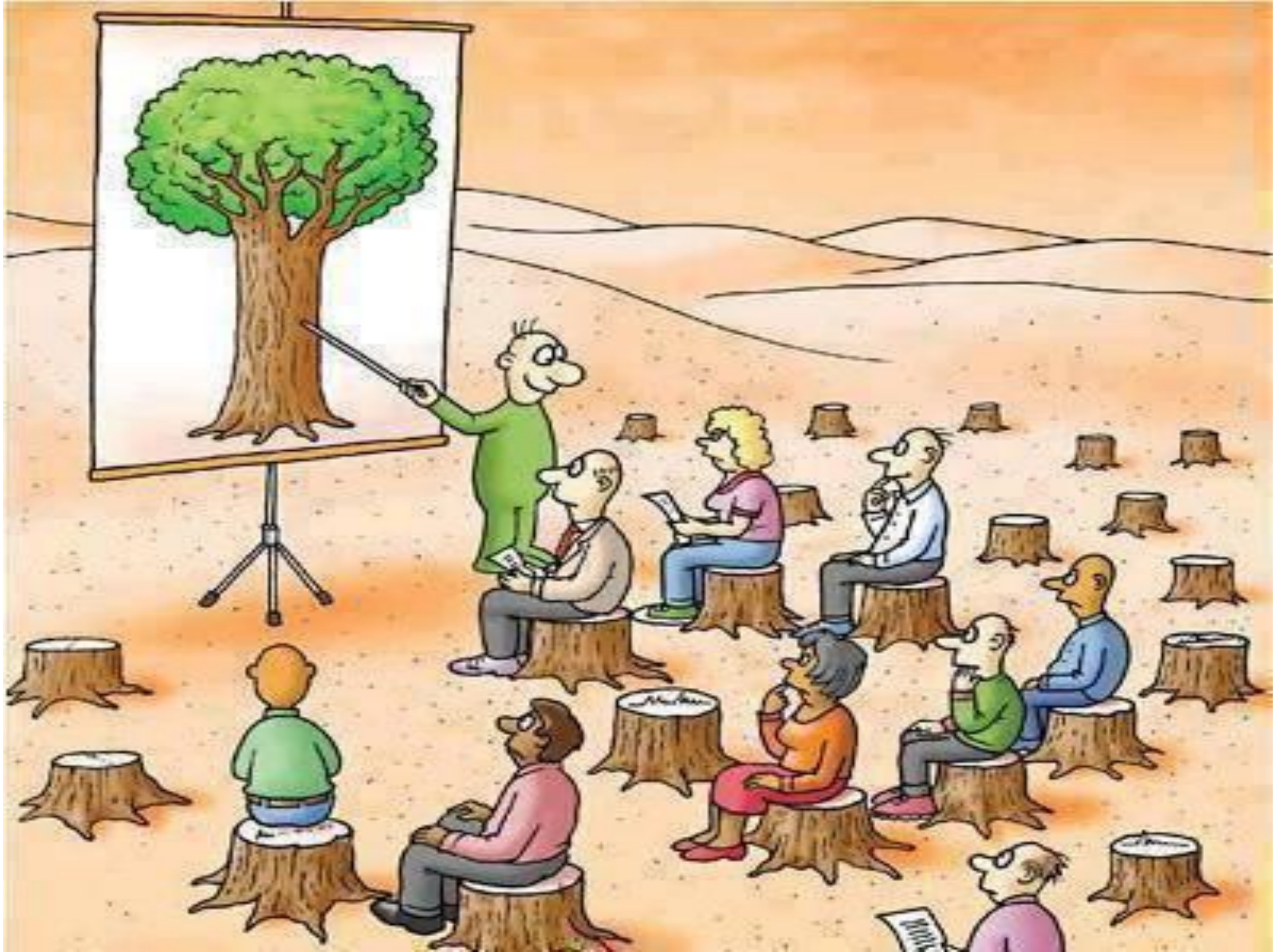


- Module 1: Introduction to Environmental Economics (12 hrs.)
- Module 2: The Design and Implementation of Environmental Policy (12 hrs.)
- Module 3: Measuring Benefits of Environmental Improvements (12 hrs.)
- Module 4: Environmental Issues (with reference to India) (4 hrs.)

INTRODUCTION TO ENVIRONMENTAL ECONOMICS

- Origin of Environmental Economics
- Meaning of Environmental Economics
- Nature of Environmental Economics
- Scope of Environmental Economics
- Significance of Environmental Economics
- Interlink between Environment and Economy
- Environmental Kuznets Curve
- Common Resources
- Tragedy of Commons
- Externalities and Property Rights
- Coase Theorem

- https://www.youtube.com/watch?v=QAEE3HN1RM0

Origin of Environmental Economics

- Environmental economics developed in its present form in the 1960s as a result of the intensification of pollution and the heightened awareness among the general public in Western countries about the environment and its importance to our existence.
- Sustainable Development Goals

- https://www.youtube.com/watch?v=m9AS6KT7a5Y

Meaning of Environmental Economics

- No longer is economics merely a science of production and distribution, it has to take into account the ecological repercussions of economic activities that could affect both production and distribution.
- It means that economics as a subject cannot exist in isolation, it cannot even be a mere study of how goods and services are produced, but at the same time it has to take into consideration the impacts of the use of resources on the environment.

"part of economics which deals with interrelationship between environment and economic development and studied the ways and means by which the former is not impaired nor the latter impeded"

- Environmental economics teaches us how to promote economic growth of nations with least environmental damage

- https://www.youtube.com/watch?v=IF9YsVpZnSE

Can you find out the differences?

If the price of apple were higher, people will buy less.

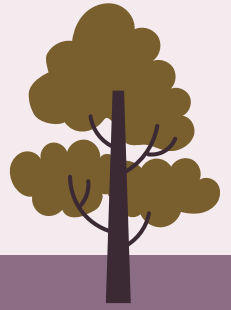
The price of gasoline will increase when OPEC meets to restrict output.

The spatial distribution of pollution emissions changes when a marketable permit system is established to regulate Sulphur emission.

People who earns large income ought to pay more income tax than people who earns low income.

The government should provide basic healthcare to all citizens.

To be fair, minimum wage should go up.



RED vs BLACK

RED		BLACK
Objective statements	VS	Subjective statemets
Cause and effect		“Should be” or “Ought to be”
Based on facts		Opinions

POSITIVE AND NORMATIVE ECONOMICS

ECONOMICS

POSITIVE

- Study of economic reality and why the economy operates as it does
- Focuses on the description and explanation of phenomena, as well as their casual relationship

NORMATIVE

- Deals with how the world ought to be.
- Focuses on what the outcome of economy or goals of public policy should be.

Nature of Environmental Economics

Positive and Normative

Micro and Macro

Env. Pollution as Eco. Problem

Social Science

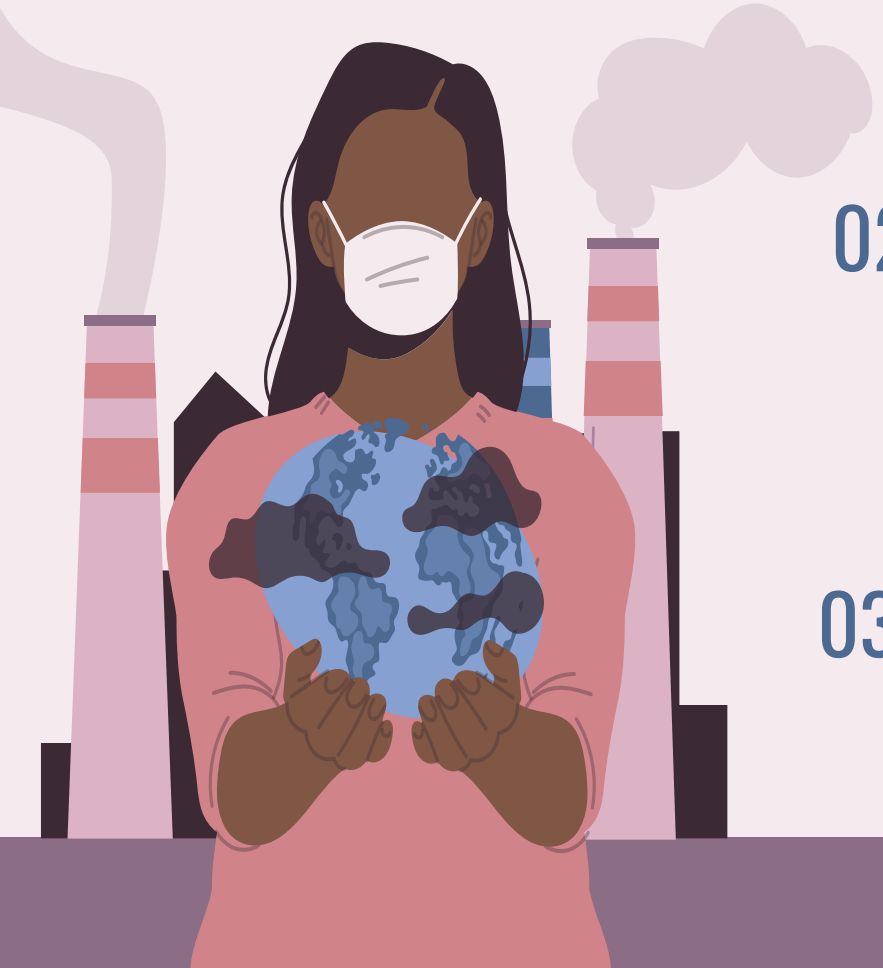
SCOPE OF ENVIRONMENTAL ECONOMICS

1. Economy-Env. analysis
2. Eco-development
3. Welfare
4. Environmental values
5. Clean Technology
6. Multi-disciplinary

Activity!

Retrospect!

Significance of Environmental Economics



01 Environmental economics will help you understand some important and controversial issues – such as climate change policy, nuclear power, recycling policy, and traffic congestion charging.

02 Environmental economics is primarily concerned with the impact of economic activities on environment and its implications for the individual firm, industry and the economy as a whole

03 The main significance of environmental economics is to maintain a balance between economic development and environmental quality

Significance of Environmental Economics

04 Environmental economics aims to tackle environmental problems from an economic welfare framework.

05 Environmental issues are about resources.

06 Presently environmental pollution is caused by misuse of existing technology and failure to develop better one

Economy and Environment!

Simon Kuznets

- Income Distribution Theory in 1955.
- Inverted U-shaped relationship between the indicators of income inequality and the level development as measured by per capita income.

Environment degradation and Economic development

Gene Grossman

Alan Krueger

Environmental Kuznet Curve

The environmental Kuznets curve suggests that economic development initially leads to a deterioration in the environment, but after a certain level of economic growth, a society begins to improve its relationship with the environment and levels of environmental degradation reduces.

Environmental Kuznet Curve

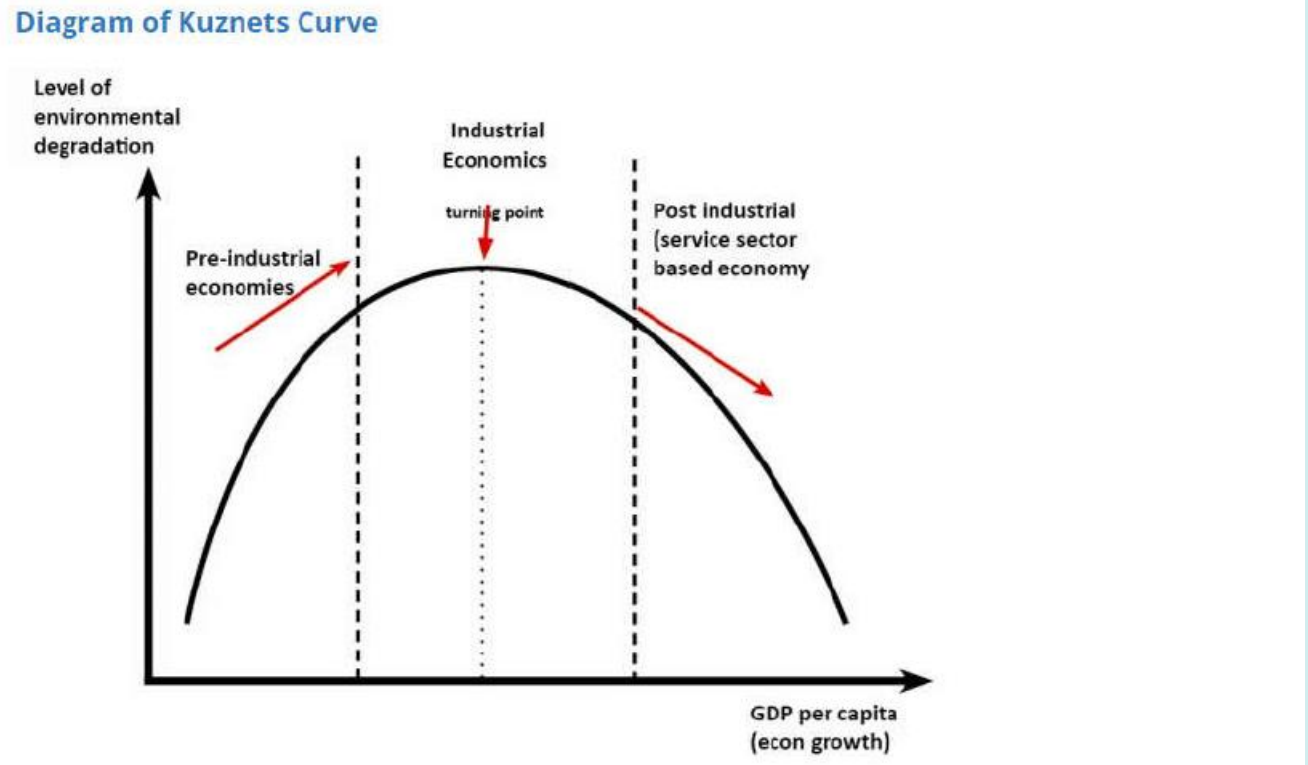


Fig 2.1 Environmental Kuznet Curve

Causes of EKC

1. Empirical evidence of declining pollution levels with economic growth.
2. Spare income with growth.
3. Focus on living standards as opposed to real GDP.
4. Role of government regulation

Causes of EKC

4. Improved technology.
5. Solar and renewable energy.
5. De-industrialisation.
6. Diminishing marginal utility of income.

Criticism of EKC

Empirical evidence is mixed

Pollution is not simply a function of income

Global pollution

Growth leads to greater resource use

Countries with the highest GDP have highest levels of CO2 emission.

The shape of the curve may be N-shaped instead of invented U Shaped

Activity!

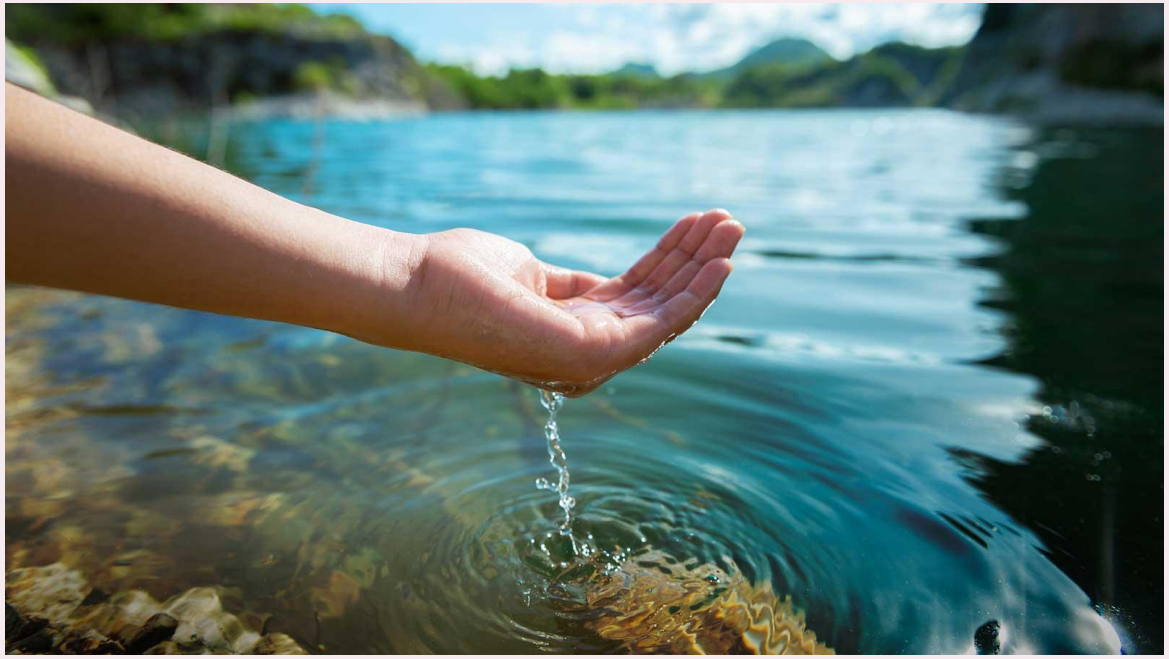
COMMON RESOURCES

“A common resource (or the "commons") is any scarce resource, such as water or pasture, that provides users with tangible benefits but which nobody in particular owns or has exclusive claim to”

Land Resources

Forest Resources

Water Resources



A major concern with common.

- **Overuse**

Overuse of common resources often leads to economic problems, such as the **Tragedy of Commons**

Tragedy of the Commons

- The tragedy of the commons is an economic theory claiming that individuals tend to exploit shared resources so that demand outweighs supply, and it becomes unavailable for the whole.
- Every individual has an incentive to consume a resource at the expense of every other individual with no way to exclude anyone from consuming.
- It results in over-consumption, under-investment, and ultimately depletion of the resource.
- As the demand for the resource overwhelms the supply, every individual who consumes an additional unit directly harms others who can no longer enjoy the benefits.
- Generally, the resource of interest is easily available to all individuals; the tragedy of the commons occurs when individuals neglect the well-being of society in the pursuit of personal gain.

“The Tragedy of commons” (1968)

- He had studied the character of herdsmen in England.
- Hardin anxiously watched out the peculiar behaviour of herdsmen that they are always prepared to add additional cattle into the pasture land in England.
- The logic prevailed that the farmer who grazed the most cattle stood to benefit most from the commons.
- But the tragedy of this kind of action is that the land was overgrazed and destroyed. This came to be known as “ Tragedy of Commons” .

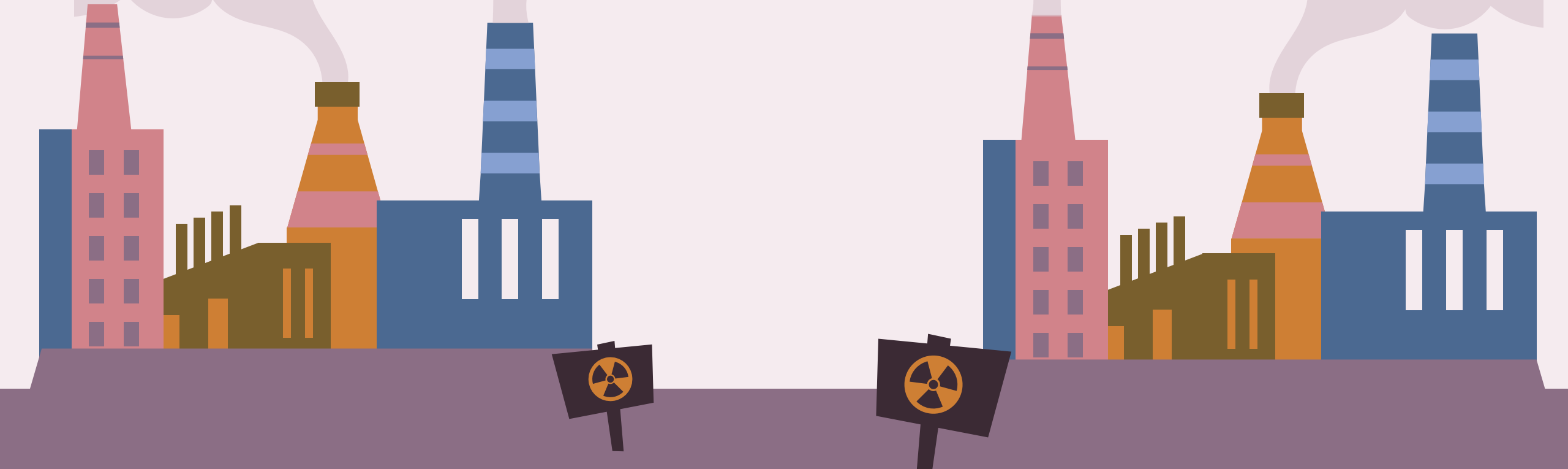
Prof. Garrett Hardin

Activity!

Externalities

Externalities occur in an economy when the production or consumption of a specific good or service impacts a third party that is not directly related to the production or consumption of that good or service.

An externality is a cost or benefit that is caused by one party but financially incurred or received by another.



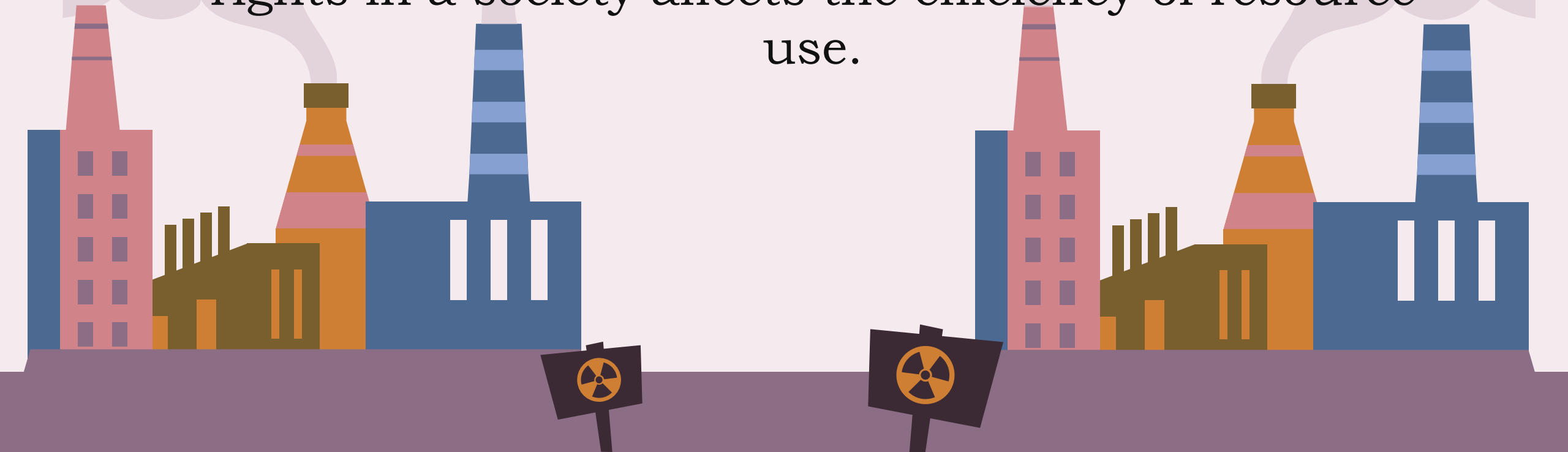
Positive and Negative

Negative externality is the indirect imposition of a cost by one party onto another.

A positive externality, on the other hand, is when one party receives an indirect benefit as a result of actions taken by another.

Property Right

Define the theoretical and legal ownership of resources and how they can be used.
In economics property rights form the basis for all market exchange, and the allocation of property rights in a society affects the efficiency of resource use.



Property Rights

- Property rights are an instrument of society and they derive their significance from the fact that they help individuals form reasonable expectations of their dealings with others.
- These expectations find expression in a society's laws, customs, and mores.
- Property rights specify how people may be benefited and harmed and therefore who must pay whom to modify the actions taken by individuals.

Property Rights and Externalities

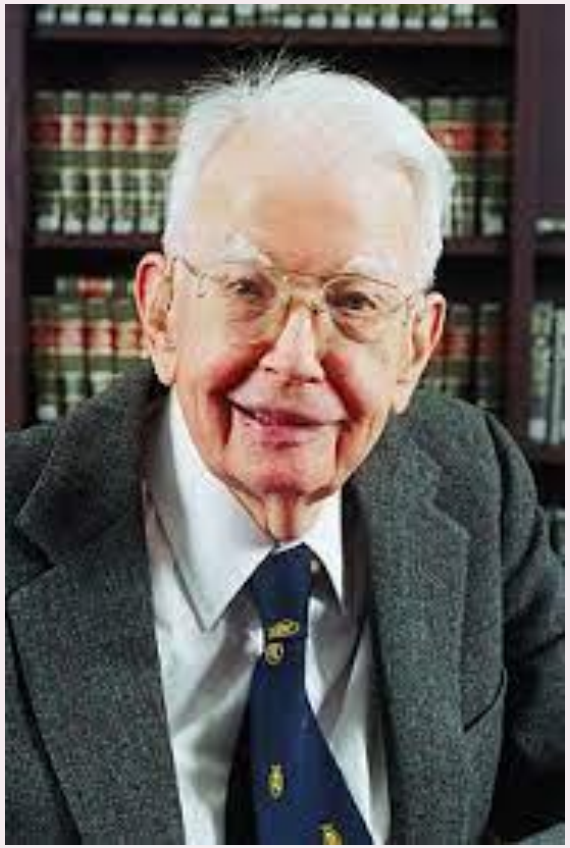
- Property rights have a close relationship with externalities.
- Property rights can help internalize externalities. However, such a process involves costs that have to be taken into account.
- Not only is it important to have a structure of property rights, it is equally important to ensure that these rights are enforced. This is where the legal system comes in.

https://www.youtube.com/watch?v=8VvqK1JRTuA

Coase Theorem

Any effort to internalize environmental externalities requires an effective scheme for assigning property rights.

Coase also believed that by assigning property rights to at least one of the parties involved (either polluter or the pollutee) there would be no effect on the final outcome of the environmental problem.



Summary

Topic Discussed

Origin of Environmental Economics; Meaning of Environmental Economics; Nature of Environmental Economics; Scope of Environmental Economics; Significance of Environmental Economics; Interlink between Environment and Economy; Environmental Kuznets Curve; Common Resources; Tragedy of Commons; Externalities and Property Rights; Coase Theorem.

Assignment

i) Understanding the interdisciplinary nature of environmental economics. (10 marks)
ii) Kuznets curve analysis. (20 Marks)

Activities

i)Understanding the link between economy and environment
ii) Common Resources; Tragedy of Commons; Externalities and Property Rights; Coase Theorem.

Suggestions!

END OF EDUCATION IS CHARACTER!